

East African Journal of Environment and Natural Resources

eajenr.eanso.org

Volume 9, Issue 3, 2026

Print ISSN: 2707-4234 | Online ISSN: 2707-4242

Title DOI: <https://doi.org/10.37284/2707-4242>



EAST AFRICAN
NATURE &
SCIENCE
ORGANIZATION

Original Article

An Analysis of Land Use and Land Cover Changes in Wenje, Tana River County, Kenya

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Article DOI: <https://doi.org/10.37284/eajenr.9.3.5506>

Date Published: **ABSTRACT**

07 August 2026

Keywords:

Remote Sensing,
Degraded
Rangelands,
Land Cover,
Rehabilitation,
Landsat Images.

Land degradation is associated with increased climate change, food insecurity, and reduced availability of livestock feed, all of which negatively affect community livelihoods. This study aimed to assess land use and land cover (LULC) changes and identify severely degraded areas in Wenje, Tana River County. Multi-temporal Landsat images from 2003, 2010, and 2017 were analyzed to detect changes in land cover. The images (Path 166, Row 61) were obtained from the United States Geological Survey (USGS) EarthExplorer and subset to extract the Area of Interest (AOI). Supervised classification using the Maximum Likelihood algorithm was performed in ArcGIS 10.7 to classify the land into six categories: water bodies, mango plantations, built-up areas, grassland, forest, and bare land. The results revealed significant land cover changes during the study period. Between 2003 and 2017, bare ground increased markedly from 8.21% to 43.95%, while forest and grassland cover declined from 21.81% to 2.69% and from 39.43% to 11.32%, respectively. Built-up areas and mango plantations expanded from 18.18% to 25.09% and from 10.05% to 15.88%, indicating increasing urbanization and agricultural development. Water bodies decreased slightly from 2.33% to 1.06%, reflecting a reduction in surface water coverage over the study. The findings demonstrate considerable land use and land cover changes in Wenje, with a marked increase in degraded land. Mapping degraded areas using satellite imagery provides valuable information for timely and targeted rangeland rehabilitation through the restoration of local grass species, afforestation, and reforestation. Such interventions can improve livestock production, enhance ecosystem resilience, and contribute to climate change mitigation. Furthermore, promoting sustainable land use practices and preventing overgrazing are essential for restoring degraded rangelands and supporting long-term livestock productivity and community livelihoods.

APA CITATION

Kiprop, A. & Koech, O. (2026). An Analysis of Land Use and Land Cover Changes in Wenje, Tana River County, Kenya. *East African Journal of Environment and Natural Resources*, 9(3), 347-360. <https://doi.org/10.37284/eajenr.9.3.5506>.

CHICAGO CITATION

Kiprop, Abel and Oscar Koech. (2026). "An Analysis of Land Use and Land Cover Changes in Wenje, Tana River County, Kenya", *East African Journal of Environment and Natural Resources* 9 (3), 347-360. <https://doi.org/10.37284/eajenr.9.3.5506>.

HARVARD CITATION

Kiprop, A. & Koech, O. (2026). "An Analysis of Land Use and Land Cover Changes in Wenje, Tana River County, Kenya", *East African Journal of Environment and Natural Resources*, 9 (3), pp. 347-360. doi: 10.37284/eajenr.9.3.5506.

IEEE CITATION

A., Kiprop & O., Koech, "An Analysis of Land Use and Land Cover Changes in Wenje, Tana River County, Kenya", *EAJENR*, vol. 9, no. 3, pp. 347-360, Aug. 2026.

MLA CITATION

Kiprop, Abel & Oscar Koech. "An Analysis of Land Use and Land Cover Changes in Wenje, Tana River County, Kenya", *East African Journal of Environment and Natural Resources*, Vol. 9, no. 3, Aug 2026, pp. 347-360, doi:10.37284/eajenr.9.3.5506.

INTRODUCTION

Land cover change is driven by several factors, including population growth, poverty, logging, and forest fires (Lambin et al., 2001). In many developing countries, shifting cultivation remains a common farming practice, where forests and bushes are cleared through burning to expand agricultural land (Allen et al., 1985; Kim et al., 2015; Veldkamp et al., 2020). This practice converts forested areas into cropland and pasture, leading to forest loss, increased landscape fragmentation, and environmental degradation (Ampofo et al., 2015).

Studies by Langat et al. (2019) reported a decline in forest cover in Tana River County due to the expansion of crop farming into arid and semi-arid areas, coupled with other human activities. These activities include the construction of upstream dams, regulated flooding, and the expansion of irrigation schemes, all of which have contributed to significant land use and land cover changes. Forests along the banks of the Tana River play a vital role in reducing soil erosion, maintaining riverbank stability, and improving water quality. However, these ecosystem services have been considerably diminished because of continued forest degradation (Glenday, 2005).

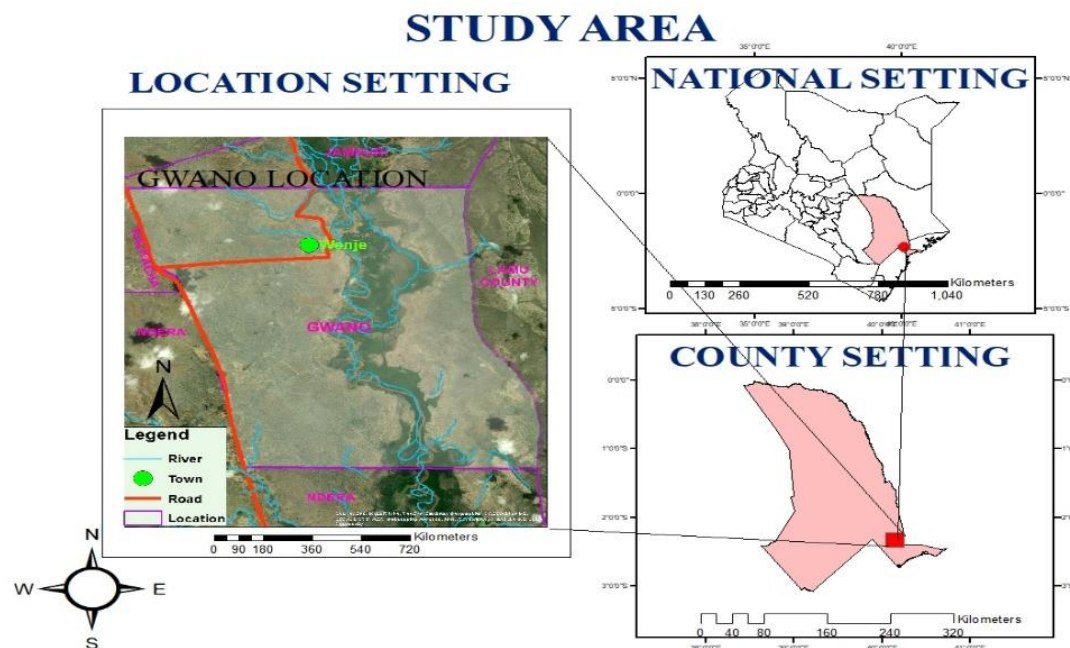
According to the Tana River County Government, the Pokomo community predominantly practices crop farming along the Tana River, while the Wardei community primarily engages in livestock production in both the riverine areas and the surrounding arid and semi-arid lands. Shifting cultivation by farming communities and overgrazing by livestock have resulted in

extensive tree clearing and increased land degradation. Despite these pressures, local communities also play an important role in the conservation and sustainable management of natural resources (Terer et al., 2004).

Several studies have demonstrated the effectiveness of satellite remote sensing in assessing vegetation and land cover changes (Tagestad, 2010; Ampofo et al., 2015; Rawat & Kumar, 2015; Thenya & Ngecu, 2017). Multi-temporal satellite imagery has been widely applied to generate change detection maps and quantify land use and land cover dynamics over time. In the present study, this approach was employed to assess land use and land cover changes in the Wenje area of Tana River County. The findings provide valuable information to support sustainable land management, rangeland rehabilitation, and informed land use planning.

MATERIALS AND METHODS**Description of the Study Area**

The study was conducted in Wenje, located in Tana River County, Kenya. A map of the study area is presented in figure 1. Tana River County is one of the six counties in Kenya's Coastal Region. It shares borders with Isiolo County to the north, Garissa County to the northeast, Kitui County to the west, Lamu County to the southeast, and Kilifi County and the Indian Ocean to the south. The county has approximately 76 km of coastline along the Indian Ocean (Government of Kenya [GOK], 2018). Geographically, Tana River County lies between approximately 2°30'S latitude and 40°20'E longitude (SMART Tana River County, 2016).

Figure 1: A Map of the Study Area Wenje in Relation to Tana River County and Kenya

Climate

Tana River County is one of Kenya's Arid and Semi-Arid Lands (ASAL) counties and is characterized by a hot, semi-arid climate. The county experiences a mean annual temperature of approximately 30°C, with the highest temperatures occurring between January and March and the lowest temperatures, averaging about 20°C, recorded between June and July. Rainfall is generally low, variable, and unreliable, ranging from approximately 280 mm to 900 mm annually. The long rainy season occurs between April and May, while the short rainy season extends from October to December, with November typically being the wettest month. These climatic conditions favor pastoralism as the dominant livelihood activity in the county (Republic of Kenya [ROK], 2018).

Soil

The soil in Tana River County is generally shallow, with depths ranging from approximately 40 to 140 cm. Sandy levees are formed through the deposition of sediments along riverbanks during the rainy season. The soil within the riverine forests is generally non-saline and non-alkaline, with good permeability and high water-holding capacity, making them suitable for

supporting diverse vegetation and agricultural activities (Maingi, 1998).

Vegetation

Grassland is the dominant vegetation type in the study area and consists of a variety of grass species, including *Echinochloa haploclada*, *Bothriochloa glabra*, and *Setaria splendida*, with the latter two being among the most common. These grass species provide important grazing resources for livestock reared by the Wardei and Orma communities.

The bushland vegetation comprises a mixture of grasses, shrubs, and woody tree species, predominantly *Acacia* species. Bushlands provide valuable forage for both wildlife and livestock, with goats and camels primarily browsing on shrubs and trees. The Orma and Wardei communities depend on these bushlands as an important source of browse for their livestock. In addition, bushlands supply firewood and other forest products that support local household livelihoods.

People and Livelihoods

Tana River County supports a diverse range of livestock species that are reared by its three main communities: the Wardei and Orma, who are

predominantly pastoralists, and the Pokomo, who primarily practice agro-pastoralism. Livestock production is a major source of food security and household income for these communities. The principal livestock products include milk, meat, hides, and skins, while livestock manure also serves as an important source of income and contributes to agricultural production (Givens et al., 2000; Lutta et al., 2021).

Approximately 40% of the county's population derives its livelihood from pastoralism, while the remaining 60% depend on crop farming, formal employment, trade, and other business activities. The Pokomo community is predominantly engaged in farming along the Tana River, whereas the Orma and Wardei communities mainly rely on pastoralism in the surrounding arid and semi-arid areas (Tana North District KIRA Report, 2013).

Data Collection

Availability Data for analysis

Multi-temporal Landsat satellite imagery was used to assess land use and land cover (LULC) changes in the study area. Landsat images acquired in 2003, 2010, and 2017 were downloaded free of charge from the United States Geological Survey (USGS) EarthExplorer. The years 2003, 2010, and 2017 were selected to provide a multi-temporal assessment of land-use and land-cover (LULC) changes over a 14-year period. These dates represent approximately seven-year intervals, which are sufficiently spaced to capture significant landscape transformations resulting from agricultural expansion, urban development, vegetation dynamics, and other anthropogenic activities. The selection of these years also ensured the availability of high-quality, cloud-free Landsat imagery with consistent spatial resolution and spectral characteristics, allowing reliable

comparison of land-cover patterns across time. Furthermore, the temporal distribution of the images enabled the evaluation of both short- and medium-term LULC trends, facilitating a comprehensive understanding of the rate, magnitude, and direction of environmental changes within the study area.

All images corresponded to Path 166 and Row 61 and were selected because they were cloud-free, ensuring high-quality image analysis. The study utilized the visible spectral bands (Bands 1, 2, and 3), representing the blue, green, and red wavelengths, respectively, for image interpretation and classification. Supervised classification using the Maximum Likelihood algorithm was performed to classify the imagery into six land cover classes: (1) water bodies, (2) mango plantations, (3) built-up areas, (4) grassland, (5) forest, and (6) bare land.

In this study, the forest class included riverine forests, forest reserves, and isolated forest patches. Grassland comprised areas dominated by grasses with scattered trees, while mango plantations referred to areas under cultivated mango trees. Bare land included exposed soil surfaces, rocky areas, and cultivated fields with little or no vegetation cover. Built-up areas consisted of settlements, livestock bomas, and other permanent structures, as described in Table 1. Landsat imagery obtained from the USGS provides a freely accessible and reliable source of data for monitoring land use and land cover changes over time (Rawat & Kumar, 2015). Satellite image interpretation enables the analysis of historical land cover patterns and facilitates the detection and mapping of changes over time, thereby supporting effective land management and planning (Mertens & Lambin, 2000). The acquisition dates of the Landsat images used in this study are presented in Table 1.

Table 1: Data Set and Characteristics

Image date of acquisition	Type of sensor	Season during time of acquisition
1. 16/12/2003	Landsat 7	Dry season
2. 01/01/2010	Landsat 8	Dry season
3. 12/01/2017	Landsat 8	Dry season

Sub Setting of the Images

Gwano Location boundary polygon obtained from the Independent Electoral and Boundaries Commission (IEBC) was used to delineate the study area from the full Landsat scene (Path 166, Row 61). Image sub setting was performed in ArcGIS using the Extract by Mask tool available in ArcToolbox, where the geo-referenced boundary polygon was used to clip the Landsat images to the Area of Interest (AOI). This process reduced the image extent to the study area, thereby improving processing efficiency and enhancing the accuracy of image classification. Image sub setting is an important preprocessing step that facilitates more accurate land use and land cover classification by eliminating unnecessary portions of the imagery (Jain et al., 1997).

Data Analysis

NDVI Analysis

The Landsat images were imported into ArcGIS 10.7 for image processing and analysis. The Raster Calculator tool, available under ArcToolbox → Spatial Analyst Tools → Map Algebra, was used to compute the Normalized Difference Vegetation Index (NDVI) for each image.

The NDVI was calculated using the following equation:

$$\text{NDVI} = (\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$$

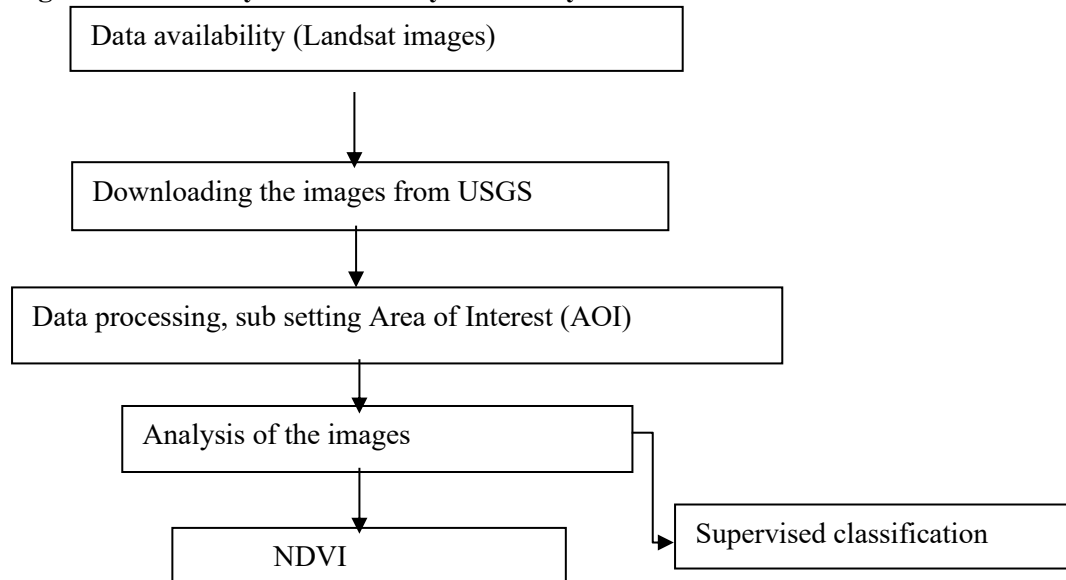
..... Equation (1)

Where **NIR** represents the reflectance in the near-infrared band and **RED** represents the reflectance

in the red band. The NDVI is based on the principle that healthy green vegetation strongly absorbs red light due to chlorophyll pigments while reflecting a large proportion of near-infrared radiation because of the internal structure of plant leaves (mesophyll). Consequently, NDVI provides a reliable measure of vegetation condition and density. NDVI analysis was performed on the Landsat images acquired in 2003, 2010, and 2017 to evaluate changes in vegetation cover over time. NDVI values range from -1 to +1, where higher positive values indicate dense, healthy vegetation, values close to zero represent sparse vegetation or bare surfaces, and negative values generally correspond to water bodies or other non-vegetated surfaces. The resulting NDVI maps were used to assess the spatial and temporal variations in vegetation cover across the study area.

Supervised Classification and Change Detection

Six land cover classes were identified: water bodies, mango plantations, built-up areas, grassland, forest, and bare land. Land cover classification was performed using the Maximum Likelihood Classification (MLC) algorithm with 70 to 150 training samples selected for each class. Changes in land cover between 2003, 2010, and 2017 were analyzed using a post-classification comparison method. All classification and analysis procedures were conducted in ArcGIS 10.7. The accuracy of each classified image was assessed to ensure the reliability and validity of the results, and the overall methodological process is illustrated in Figure 2.

Figure 2: Summary of Availability and Analysis of Data**Table 2: Land Cover Classification Scheme Used in the Study Area**

Land cover type		Description
1	Forest Vegetation	River line forest, forest reserve and patch of the forest
2	Water	Dams, rivers, and wetlands
3	Grassland	Areas covered with grass and trees
4	Built up areas	Settlement, buildings and livestock shade
5	Bare Land	Exposed are with no vegetation cover, areas covered by rocks
6	Mango plantation	Mango trees

RESULTS

The land use and land cover classification results for 2003, 2010, and 2017 are presented in Figures 3 and 4 and summarized in Table 3. The results indicate considerable changes in land cover composition over the study period.

Water bodies occupied 481.32 ha (2.33%) of the study area in 2003. This decreased to 276.75 ha (1.33%) in 2010 and further declined to 219.24 ha (1.06%) in 2017. Mango plantations covered 2,076.18 ha (10.05%) in 2003, increased slightly to 2,112.75 ha (10.22%) in 2010, and expanded further to 3,284.64 ha (15.88%) in 2017.

Bare land exhibited the greatest increase during the study period. It covered 1,696.95 ha (8.21%) in 2003, increased to 1,995.75 ha (9.65%) in 2010, and expanded substantially to 9,090.18 ha (43.95%) in 2017. Similarly, built-up areas

increased from 3,760.00 ha (18.18%) in 2003 to 4,837.23 ha (23.39%) in 2010 and reached 5,189.85 ha (25.09%) in 2017.

Grassland was the dominant land-cover class in 2003, occupying 8,154.18 ha (39.43%) of the total area. However, its coverage declined to 6,524.73 ha (31.55%) in 2010 and further decreased to 2,341.26 ha (11.32%) in 2017. Forest cover accounted for 4,509.72 ha (21.81%) in 2003 and increased slightly to 4,934.25 ha (23.86%) in 2010 before declining markedly to 556.29 ha (2.69%) in 2017.

Overall, the results show substantial land use and land cover changes between 2003 and 2017, characterized by increases in bare land, built-up areas, and mango plantations, and decreases in grassland, forest cover, and water bodies. This indicates considerable landscape transformation within the study area over the 14-year period.

Figure 3: Land Cover and Land Use in Ha

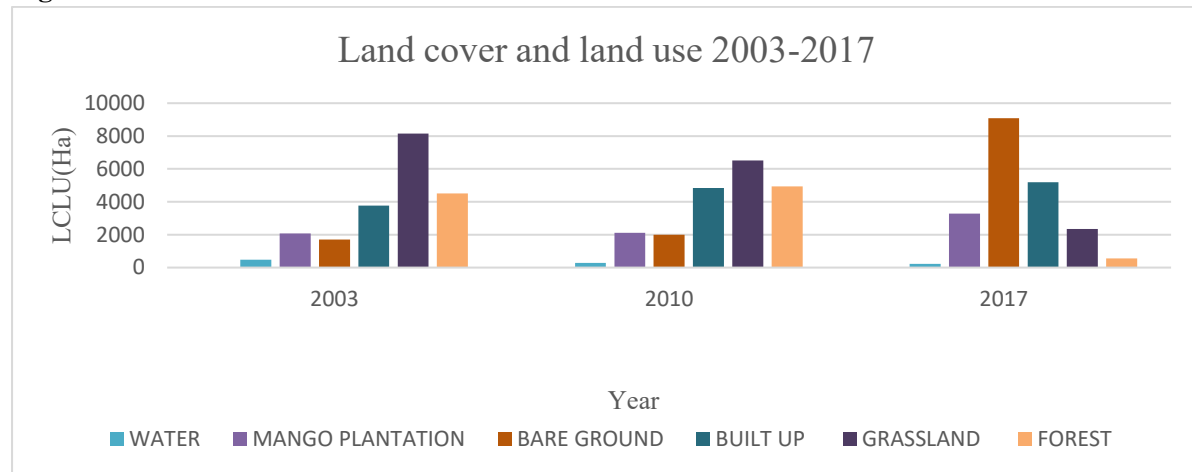


Figure 4: Land Cover Changes for the Year 2003, 2010 and 2017

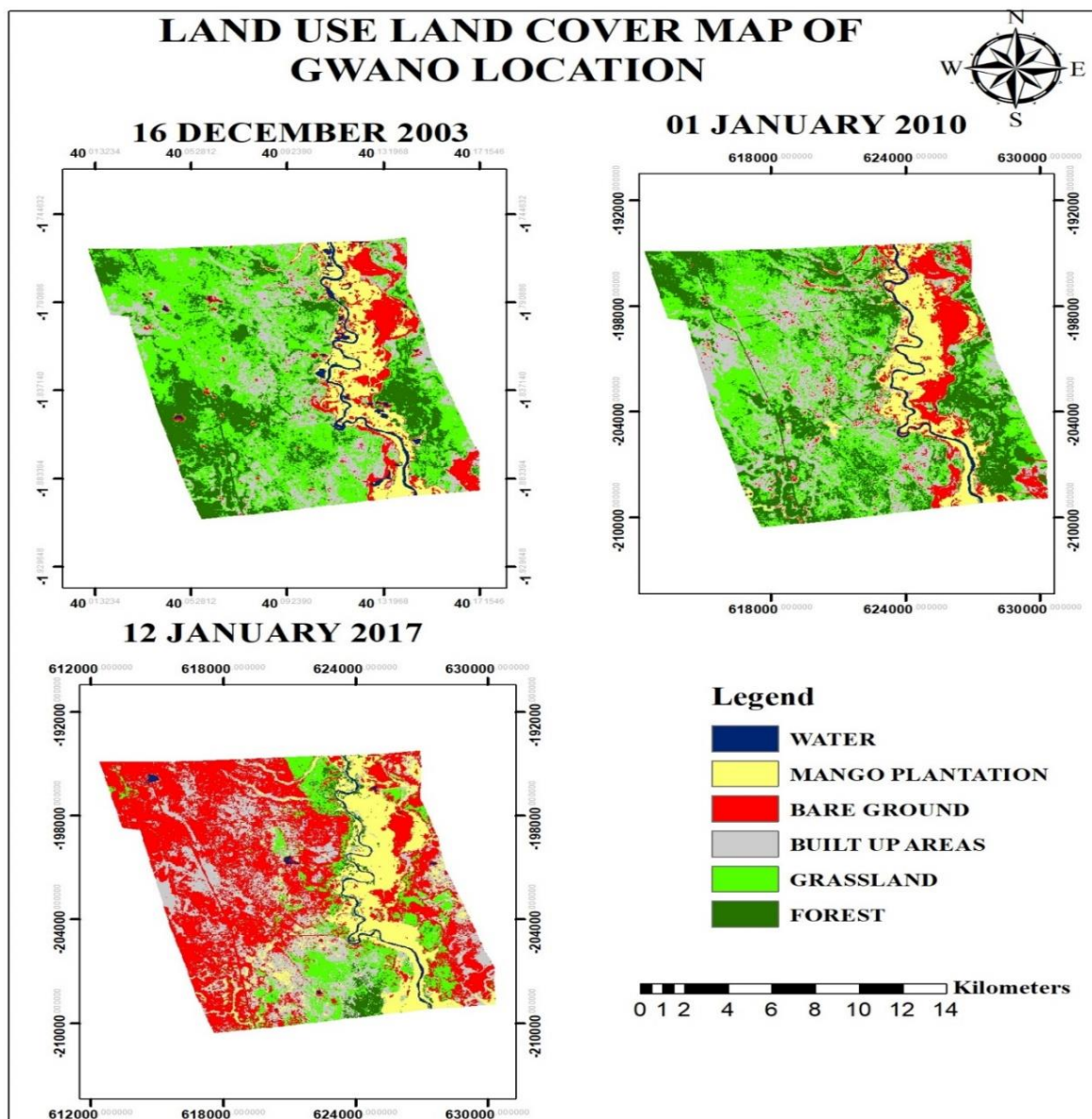


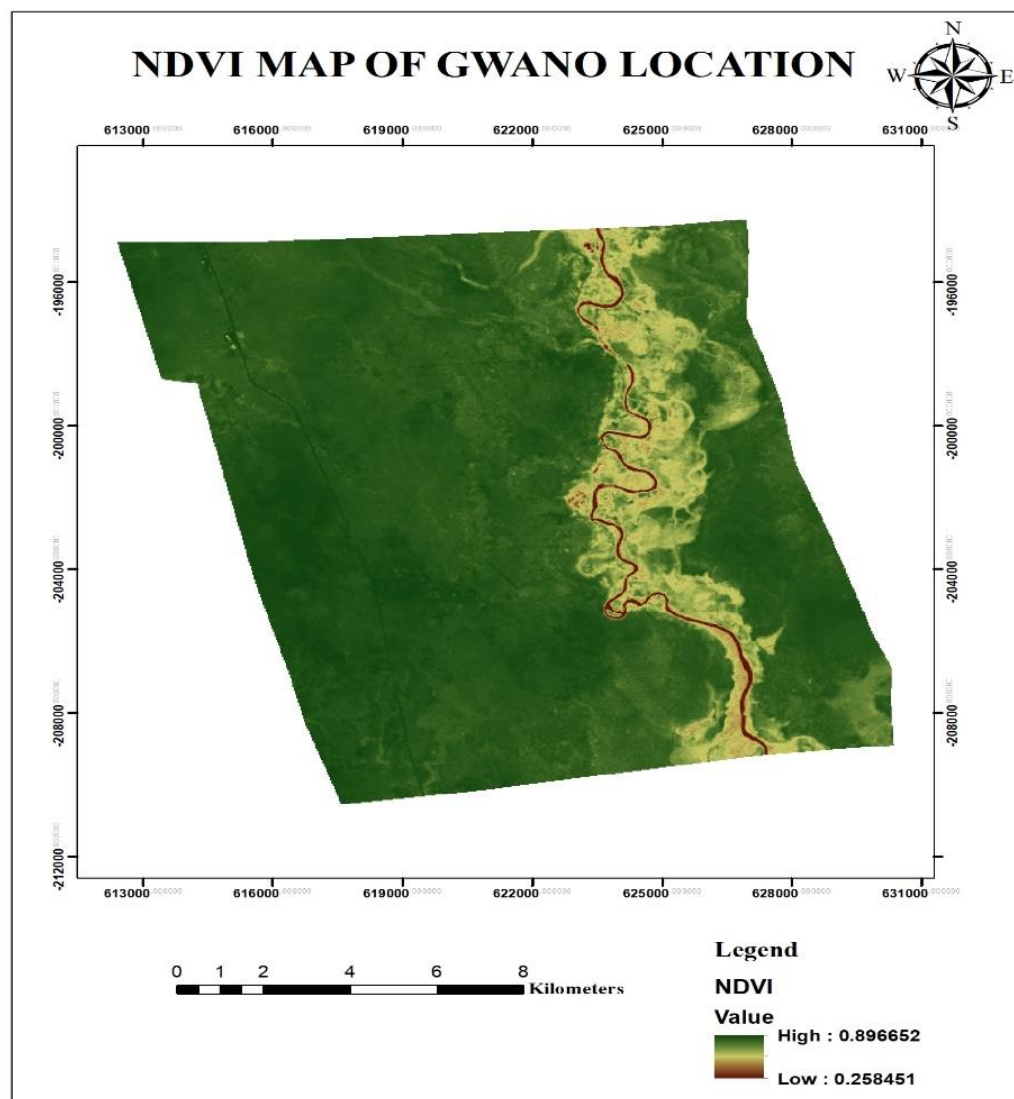
Table 3: Area and Percentage Change of Land Use Land Cover in Wenje

	2003	2003	2010	2010	2017	2017
CLASSES	Area in Ha	% Area	Area in Ha	% Area	Area in Ha	% Area
WATER	481.32	2.33	276.75	1.34	219.24	1.06
MANGO PLANTATION	2,079.18	10.05	2,112.75	10.22	3,284.64	15.88
BARE GROUND	1,696.95	8.21	1,995.75	9.65	9,090.18	43.95
BUILT UP	3,760.11	18.18	4,837.23	23.39	5,189.85	25.09
GRASSLAND	8,154.18	39.43	6,524.73	31.55	2,341.26	11.32
FOREST	4,509.72	21.81	4934.25	23.86	556.29	2.69
TOTAL AREA	20681.46	100	20681.46	100	20681.46	100

NDVI Analysis Showing How Vegetation Cover has Change over a Period of 20 Years in Wenje, Tana River County

According to the NDVI results, most of the study area is characterized by healthy vegetation, with

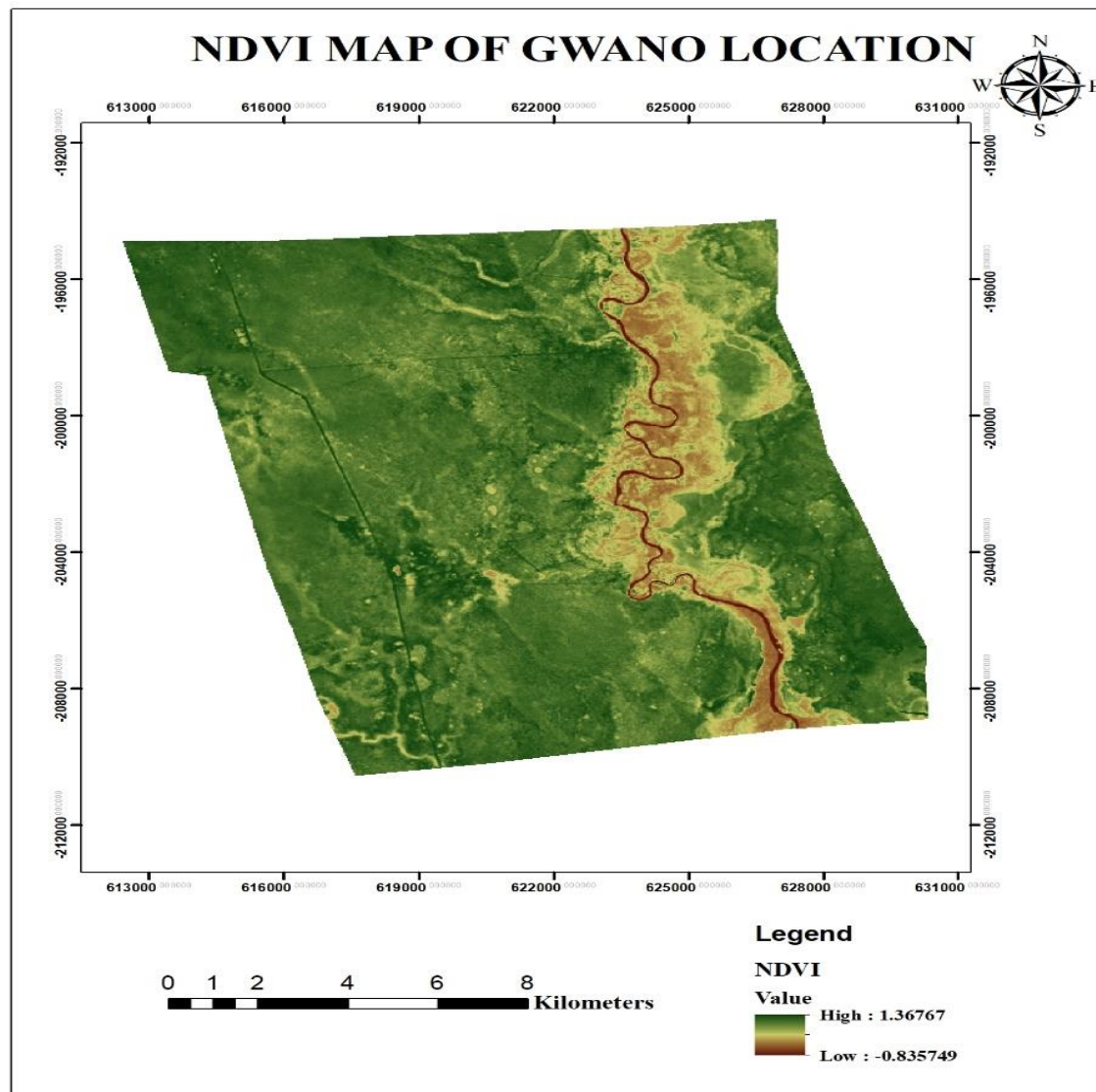
high NDVI values reaching up to 0.86. In contrast, degraded areas are indicated by low NDVI values of approximately 0.26, reflecting sparse or stressed vegetation cover as shown in Figure 5.

Figure 5: NDVI for the Year 2003

The NDVI values ranged from -0.84 to 1.37 across the study area. The results indicate the presence of generally healthy vegetation; however, some areas recorded low NDVI values of approximately 0.1 , reflecting sparse or

degraded vegetation cover. Overall, vegetation health has declined compared to the year 2003, indicating a reduction in vegetative vigor over time as shown in Figure 6.

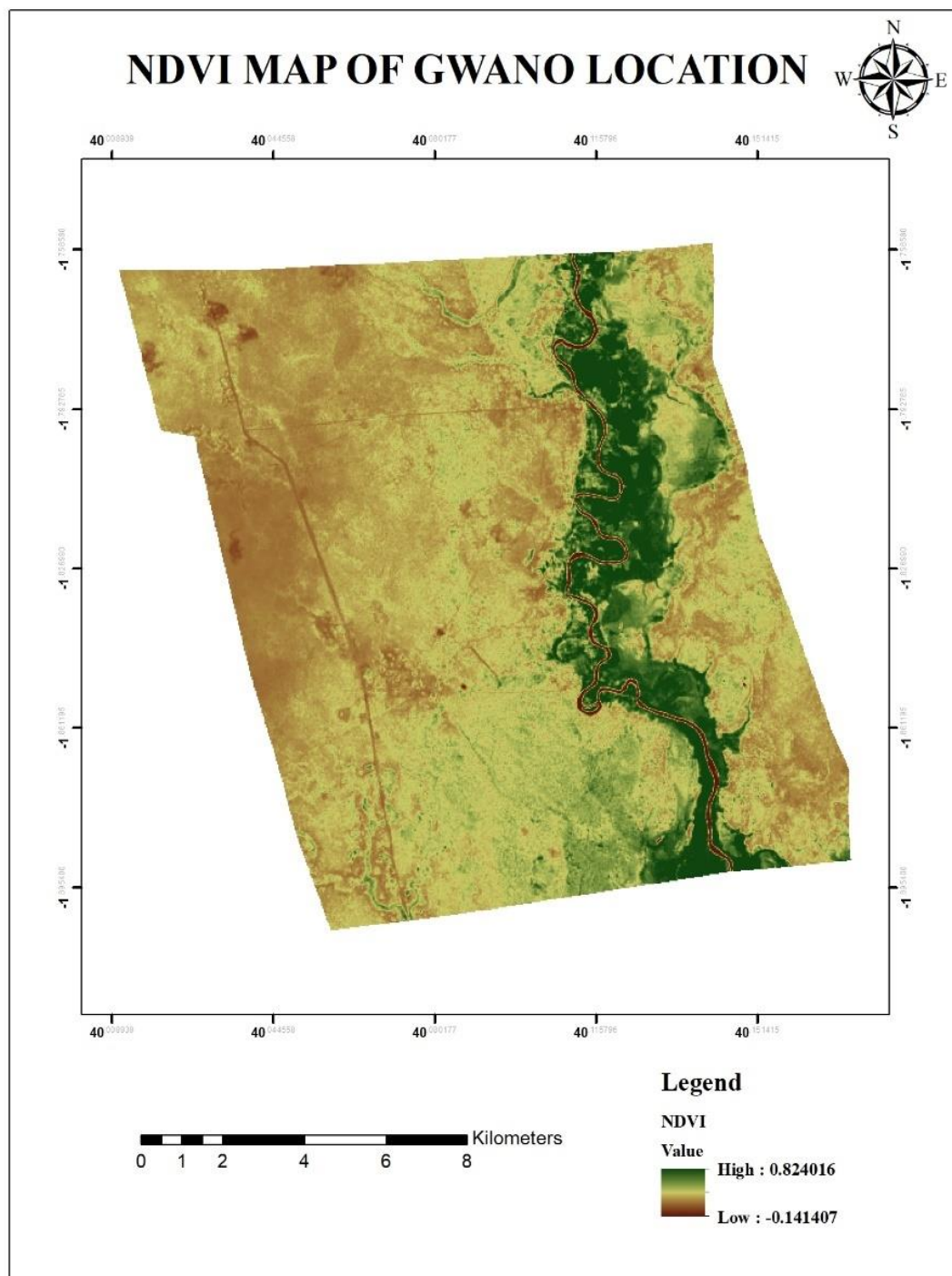
Figure 6: NDVI for the Year 2010



The NDVI values in 2017 ranged from -0.14 to 0.82 . The results indicate that vegetation cover in 2017 had declined compared to 2003 and 2010. This reduction in NDVI values reflects a

deterioration in vegetation health and indicates that land degradation had become more pronounced in 2017 as shown in Figure 7.

Figure 7: NDVI for the Year 2017



DISCUSSION

Grasslands play a critical role in supporting livestock production and maintaining ecosystem functions in arid and semi-arid lands (Lüscher et al., 2014). The land-cover analysis revealed a substantial decline in grassland cover between 2003 and 2017, accompanied by significant increases in built-up areas and bare land. This finding suggests a progressive degradation of

grazing resources within the study area, consistent with the study objective of assessing land-use and land-cover dynamics over time. The reduction in grasslands is likely associated with a combination of overgrazing, increasing human population, and recurrent drought events that limit vegetation regeneration. Similar trends have been reported in the Tana River Basin and other dryland ecosystems of Kenya, where increasing grazing

pressure and climatic variability have led to declining vegetation cover and rangeland productivity (Benoit & Askins, 2002; Kipkemboi et al., 2017). The observed decline in grasslands has important implications for livestock production, ecosystem resilience, and climate change mitigation, given the role of grasslands in carbon sequestration and forage provision (Laidlaw & Frame, 2013).

The classified images also showed a marked increase in bare land, which rose from 8.21% in 2003 to 43.97% in 2017. This substantial increase indicates accelerated land degradation and reduced vegetation cover over the study period. The expansion of bare land can be attributed to prolonged drought conditions and unsustainable grazing practices that expose soil surfaces and reduce vegetation recovery. This interpretation agrees with findings by Benoit & Askins, (2002) and Kipkemboi et al., (2017), who reported that overgrazing and recurrent droughts in the Tana River region contributed to vegetation loss, biodiversity decline, and land degradation. The increase in bare land observed in this study therefore reflects the growing vulnerability of rangeland ecosystems to both anthropogenic and climatic pressures.

A decline in water bodies was also observed during the study period, suggesting changes in hydrological conditions within the basin. The reduction may be associated with decreased rainfall in both the study area and upstream catchments, including the Aberdare Range, as well as increasing abstraction of river water for irrigation (Langat et al., 2019). The area supports extensive irrigated agriculture, including maize, rice, onions, vegetables, tomatoes, and watermelon production (Hussain & Baker, 2016), which may further reduce surface water availability. These findings are consistent with those reported by Langat et al. (2019), who documented declining water resources and increasing pressure on riverine ecosystems within the Tana River Basin. The observed reduction in water bodies may have significant implications for agricultural production, livestock watering, and the ecological integrity of aquatic habitats.

The study further revealed a decline in forest cover between 2003 and 2017. This trend suggests continued conversion of natural vegetation to agricultural land and settlements, particularly along riverine areas where farming activities are concentrated. Forest loss in the study area is likely driven by agricultural expansion, charcoal production, logging, and timber extraction. Similar observations have been reported by Mbora & Meikle (2004) and Langat et al., (2019), who identified agricultural encroachment and unsustainable resource utilization as major drivers of riverine forest degradation in the Tana River Basin. The decline in forest cover is of concern because forests provide essential ecosystem services, including carbon sequestration, biodiversity conservation, climate regulation, and livelihood support through the provision of timber, fuelwood, fodder, medicinal products, and non-timber forest products (Krieger, 2001; Hudiburg et al., 2011; Betts et al., 2019).

Conversely, the analysis indicated an increase in mango plantations and built-up areas during the study period. The expansion of mango farming reflects increasing commercialization of agriculture and growing market demand for fruit production in Hola and neighbouring urban centres. Likewise, the growth of built-up areas is indicative of population increase, settlement expansion, and changing socio-economic conditions that heighten demand for housing and infrastructure. These findings support previous studies that have linked population growth and agricultural intensification to land-cover transformation in the Tana River Basin (Mbora & Meikle, 2004; Langat et al., 2019). While these changes may contribute to local economic development, they also exert pressure on natural ecosystems through continued conversion of grasslands and forests.

Overall, the results demonstrate significant land-use and land-cover transformations between 2003 and 2017, characterized by declining grassland, forest, and water-body cover and increasing bare land, agricultural land, and settlement areas. These changes indicate growing environmental pressure resulting from human activities and

climatic variability. The findings are consistent with previous studies conducted in the Tana River Basin and highlight the need for integrated land management strategies that promote sustainable grazing, improved water resource management, afforestation, and restoration of degraded lands. Such interventions are essential for enhancing ecosystem resilience, sustaining livelihoods, and reducing further environmental degradation in the study areas.

CONCLUSIONS AND RECOMMENDATIONS

This study demonstrated the effectiveness of multi-temporal Landsat imagery and change detection techniques in assessing land use and land cover dynamics and identifying degraded areas within the study area between 2003 and 2017. The results revealed substantial land cover changes over the study period, characterized by a decline in grassland, forest cover, and water bodies, alongside an increase in bare land, built-up areas, and mango plantations.

The increase in bare land from 8.21% in 2003 to 43.97% in 2017 indicates extensive land degradation and a reduction in vegetation cover. The NDVI analysis further confirmed a general decline in vegetation condition and health over time. These findings suggest that overgrazing, recurrent droughts, agricultural expansion, and settlement growth are important drivers of land cover change in the area. The reduction in grassland and water resources is likely to have negative implications for livestock production and pastoral livelihoods, given the dependence of local communities on rangeland resources for forage and water.

The observed expansion of built-up areas and mango plantations reflects increasing human influence on land resources and changing land-use practices. While these changes may contribute to local economic development, they also increase pressure on natural ecosystems and accelerate the conversion of vegetated land covers.

Overall, the findings indicate that the study area is undergoing increasing environmental degradation

and landscape transformation. To reverse these trends, sustainable land management strategies should be promoted, including controlled grazing, restoration of degraded rangelands, afforestation and reforestation initiatives, and the reseeding of degraded areas with indigenous grass species. Such interventions would contribute to improved ecosystem resilience, sustainable livestock production, and long-term conservation of natural resources.

Competing Interests

The authors declare that there are no competing interests associated with this study.

Acknowledgements

The author expresses sincere gratitude to the Tana River Beef Project (TARIBE) for funding this research. Special thanks are extended to Mr. Abdalla Omar and his family for their invaluable support and assistance during the data collection process.

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